crystal potential

$$\hat{\mathcal{V}}_{Ne} = -\sum_{i,j} \frac{e^2}{4\pi\epsilon_0} \frac{Z_j}{|\mathbf{r}_i - \mathbf{R}_j|}$$

guess of initial electron density $\rho_0(\mathbf{r})$

effective KS pontential $V_{KS}[\rho(\mathbf{r})]$

solving the KS equation

$$\left[\frac{1}{2}\nabla^2 + V_{KS}(\mathbf{r})\right] \phi_i^{KS}(\mathbf{r}) = E_i \phi_i^{KS}(\mathbf{r})$$

electron density

$$\rho(\mathbf{r}) = \sum_{i=1}^{N} \left|\phi_i^{KS}(\mathbf{r})\right|^2$$

converged?

no

yes

energy, forces, band structures
magnetic structures
optical properties, etc

